Efficient Retrieval for Small Language Models

Abstract: We study whether small retrieval-augmented models can match larger models on question answering benchmarks.

Method: We combine a lightweight dense retriever with a 1B parameter generator and evaluate on NaturalQuestions.

Results: Our approach reaches 62.3 EM on NaturalQuestions, within 2 points of a 7B parameter baseline while using 6x less compute.

Limitations: We only evaluate on English text and do not test multi-hop reasoning benchmarks.